

★ WEEK 1 · LECTURE

The World of ML & the Finance Problem

What machine learning is — explained from zero

Expanded edition · 38 slides · with three hands-on labs

No background needed. We start with a true story, use everyday analogies, and build — one small idea at a time — the foundation of everything that follows.

Where We Are Going Today

Four simple questions — by the end you will run real Python in your browser

1

What even IS machine learning?

Writing a recipe yourself vs. letting a chef taste and learn.

2

What are the three kinds of learning?

Three flavors — supervised, unsupervised, reinforcement.

3

Why do models go wrong? (bias & variance)

A dartboard and a memorizing student make it crystal clear.

4

Why is FINANCE so hard for ML?

A famous 1998 disaster shows the answer — and our response.

1

PART 1 · FOUNDATIONS

What Is Machine Learning?

Start here — the single most important idea

A Question to Start

How would YOU teach a computer to spot a cat in a photo?

You could try writing rules: "a cat has two pointy ears, whiskers, fur..." But a folded-ear cat? A cat seen from behind? A drawing of a cat? Rules break instantly.

The old way breaks down

- Writing rules by hand works for simple things.
- But the real world has endless exceptions.
- You can never write enough rules to cover them all.

The machine-learning way

- Instead, show the computer 10,000 photos already labeled 'cat' or 'not cat'.
- Let it discover the pattern on its own.
- This is the heart of machine learning.

Old Way vs. ML — A Cooking Story

Machine learning = learning the recipe from examples, instead of being told the recipe.

Old Way vs. Machine Learning — A Cooking Analogy

Old Way: You Write the Recipe

A human writes every rule by hand.

"IF debt is high AND income is low,
THEN reject the loan."

You + the computer follow the recipe
exactly, step by step.

**Problem: real markets have millions
of patterns — too many to write down.**



ML: The Computer Tastes & Learns

**You show many finished dishes +
whether each was tasty.**

The computer figures out the recipe
by itself, from the examples.

Show 10,000 past loans + who repaid;
it learns the pattern automatically.

**Win: it finds patterns no human could
ever write down by hand.**

So, What Is Machine Learning?

One sentence — then we never make it complicated again

Machine learning is teaching a computer to find patterns in data, so it can make good guesses about new things it has never seen.

Three words you'll hear forever

- Data: the examples we show the computer (e.g. past loans).
- Pattern: the hidden rule the computer discovers.
- Prediction: its best guess about something new.

Why it matters in finance

- Markets create oceans of data every second.
- Hidden patterns can mean profit — or danger to avoid.
- No human can read it all; a machine can help.

Rules vs. Learning — Two Ways to Build Software

The same contrast, laid side by side — keep this table in your head all semester

	Hand-written rules	Machine learning
Who finds the pattern?	A human writes every if-then rule.	The computer discovers it from examples.
Exceptions	Each one needs a new rule — the list explodes.	Handled naturally, if the examples cover them.
When the world changes	Someone must notice and rewrite the code.	Retrain on fresh data — the pattern updates.
Finance example	'Flag any transfer over \$5,000.'	Learn what fraud looks like from labeled cases.

Neither is 'better' everywhere — simple rules still win when data is scarce and stakes are clear.

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PART 2 · THE THREE FLAVORS

Three Kinds of Learning

Supervised, unsupervised, reinforcement

The Three Flavors of Machine Learning

Almost every ML method is one of these three — we focus mostly on supervised learning.

Three Paradigms of Machine Learning

1 · Supervised

Learn from labeled data

Input X with known answer y .
Model learns the mapping X to y .

Finance examples:

- Default prediction (yes / no)
- Stock direction (up / down)
- Credit scoring

2 · Unsupervised

Find structure, no labels

Only input X is given.
Model finds hidden patterns.

Finance examples:

- Customer segmentation
- Anomaly / fraud detection
- Factor extraction (PCA)

3 · Reinforcement

Learn by trial and reward

Agent acts in an environment.
Rewards guide better actions.

Finance examples:

- Optimal trade execution
- Dynamic portfolio rebalancing
- Sequential recommendation

Supervised Learning = Learning with an Answer Key

Like a teacher grading flashcards until you get them right

Think of flashcards: front = a question, back = the correct answer. Show the computer thousands of cards; it learns to answer new ones.

In ML words

- Features (X): the front of the card — the clues.
- Label / target (y): the back of the card — the answer.
- The model learns the link from clues to answer.

Finance example

- Front (X): a borrower's income, debts, history.
- Back (y): did they repay? (yes / no)
- Goal: judge a brand-new applicant correctly.

The Same Idea, as a Tiny Formula

Don't panic — it just says 'answer = a function of the clues': y-hat equals f applied to x

$$\hat{y} = f(x_1, x_2, \dots, x_d)$$

Piece by piece

- y-hat (the hat means 'estimate'): the computer's guess.
- $x_1, x_2, \dots, x_d$: the clues / features (d of them).
- f: the learned pattern — ML's whole job is finding a good f.

A concrete f

- Suppose f says: 'risk = 2 x debt – 1 x income'.
- Plug in a person's debt and income, out comes a risk score.
- Training = finding the right numbers (2 and –1) from past data.

Two Jobs for Supervised Learning

Regression predicts a number; classification predicts a category — that is the difference

Regression — predict a NUMBER

'How much?'

The answer y is a quantity on a scale. The model outputs its best numeric estimate.

tomorrow's return · a house price · next-quarter earnings

Classification — predict a CATEGORY

'Which kind?'

The answer y is one of a few boxes. The model outputs a box (often with a probability).

default / no default · fraud / not fraud · buy / hold / sell

Same formula $\hat{y} = f(x)$ in both cases — only the type of answer changes.

Unsupervised Learning = Sorting with No Answer Key

Like tidying a messy room into piles that 'go together'

Imagine a pile of mixed Lego with no instructions. You naturally group them: big bricks here, wheels there, flat pieces over there.

What the computer does

- It gets data with NO labels (no answer key).
- It finds natural groups or hidden structure.
- Nobody told it the groups — it discovers them.

Finance examples

- Group customers by spending habits → better product offers.
- Spot a weird transaction that fits no group → possible fraud.
- Boil 500 stocks down to a few key 'themes' (factors).

Reinforcement = Learning by Trial and Reward

Like training a puppy, or getting better at a video game

Play a game: good moves earn points, bad moves lose them. Over many tries, you learn the moves that win — without anyone handing you a manual.

The loop

- The agent takes an action in an environment.
- It gets a reward (good) or penalty (bad).
- It slowly learns the actions that earn the most reward.

Finance examples

- Buy a huge stock position slowly, to avoid moving the price (Week 13).
- Rebalance a portfolio as markets change.
- Decide which product to suggest a customer next.

Which Flavor Is It? — A 60-Second Drill

Cover the right column, answer out loud, then check yourself

Finance task	Which flavor?
Predict next-quarter earnings from past financials	Supervised — regression (the answer is a number)
Group 500 stocks into a few style clusters	Unsupervised — no labels, find structure
Learn to slice a big order to minimize market impact	Reinforcement — actions, rewards, trial and error
Flag a transaction unlike anything seen before	Unsupervised — anomaly detection
Approve or deny a loan using labeled repayment history	Supervised — classification (yes / no)

If you got 4 of 5, you own the three flavors already.

Three Kinds — One-Line Memory Hooks

If you remember only three pictures — flashcards, Lego piles, a video game — you've got it

1

Supervised = learning with an answer key

Flashcards. We have the labels. → Most of this course (default, fraud, prices).

2

Unsupervised = sorting with no answer key

Tidying Lego into piles. No labels. → Customer groups, fraud anomalies, factors.

3

Reinforcement = learning by trial & reward

Training a puppy / playing a game. → Trade execution, dynamic recommendation.

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PART 3 · BIAS & VARIANCE

Why Models Go Wrong

The bias-variance idea — the heart of all ML

A Student Who Memorized the Answers

Two students study for a math test — which one really learned?

Student A — the memorizer

- Memorizes every homework answer word for word.
- Perfect score on homework.
- On the real test (new questions) → fails badly.
- This is OVERFITTING.

Student B — understands

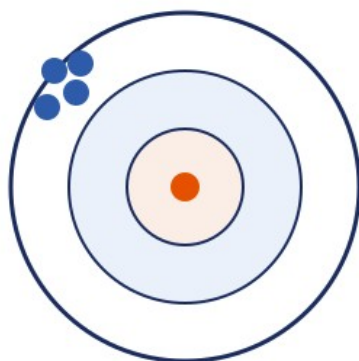
- Learns the underlying method, not the exact answers.
- Slightly imperfect on homework.
- On the real test → does well.
- This is what we WANT.

A memorizing model is Student A — it looks great in practice and fails for real.

Bias vs. Variance — The Dartboard

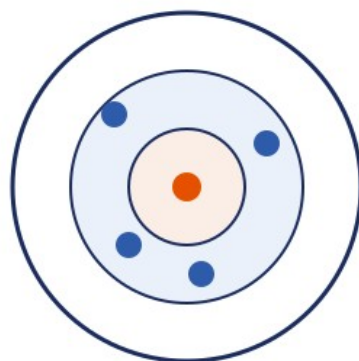
Bias = consistently off-target. Variance = all over the place. We want neither.

Bias vs. Variance — Think of Throwing Darts



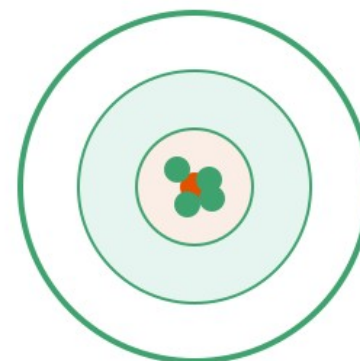
High Bias

Tight cluster, but
far from bullseye
(model too simple)



High Variance

Around the center, but
scattered everywhere
(model too sensitive)



Just Right

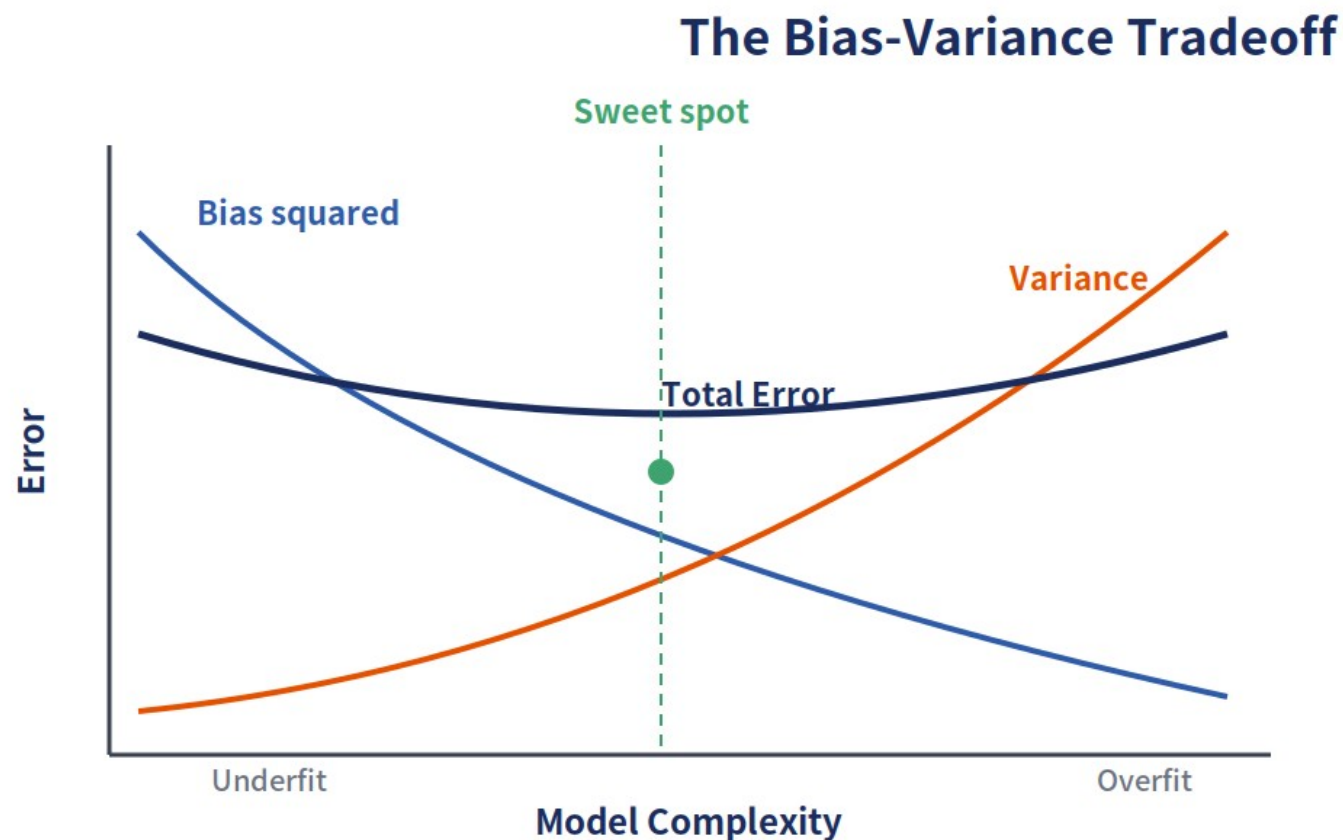
Tight AND centered
on the bullseye
(the goal!)

**The bullseye =
the true answer**

Each dart =
one prediction from
a re-trained model.

The Trade-off, as a Picture

As a model gets more complex: bias falls, but variance rises. The sweet spot is in the middle.



How to read it

Bias = error from a model too simple (misses the truth).

Variance = error from a model too sensitive to its data.

Goal: balance the two to minimize total error.

In finance, low signal makes overfitting especially dangerous.

The One Formula Worth Remembering

Total mistake = three parts added together

$$\text{MSE} = \underbrace{\text{Bias}^2}_{\text{systematic error}} + \underbrace{\text{Variance}}_{\text{sensitivity to data}} + \underbrace{\sigma_{\varepsilon}^2}_{\text{irreducible}}$$

In plain English

- Bias²: how far off on average (Student A on new questions).
- Variance: how much answers jump when data changes (scattered darts).
- Noise: pure randomness no one can predict — the market's coin-flips.

Why finance is brutal here

- Financial data is mostly noise with a tiny bit of signal.
- A fancy model easily 'learns' the noise → overfitting.
- That's why simple models often win in finance.

Underfitting vs. Overfitting — Symptoms and Cures

Read your train and test errors like a doctor reads a chart

Signal	Underfitting (too simple)	Overfitting (too complex)
Train error	High — misses even the examples it saw.	Near zero — it memorized them.
Test error	High — about as bad as training.	Much worse than training — the gap is the tell.
Dartboard picture	Tight cluster, far from the bullseye (bias).	Darts scattered everywhere (variance).
Typical cure	Add features, use a richer model.	Simplify, regularize, get more data (Week 4).

Memorize the second row: a large train-test gap is the fingerprint of overfitting.

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PART 4 · THE FINANCE PROBLEM

Why Finance Is So Hard

A true story, and the lesson it teaches

A Tale of Two Funds (1998)

Nobel Prize winners lost billions; a quiet code-breaker won for decades

"Same mathematics. Opposite fates. The difference was not brains — it was process."

LTCM — the genius trap

- Run by Nobel Prize-winning economists.
- Bet borrowed money (huge leverage) on ONE big idea.
- In 1998 the single bet collapsed — about \$4.6 billion lost in months.
- Lesson: brilliance + one fragile bet = ruin.

Renaissance — the machine

- Run by mathematicians and code-breakers.
- Combined thousands of tiny, weak signals.
- Used a careful, checked pipeline — not one hero bet.
- Delivered famously strong returns for decades.

The Real Lesson: Process Beats Genius

López de Prado's two paths: a production line of small steps, not one fragile masterpiece.

Why Financial ML Projects Fail: Two Paradigms

Sisyphus Paradigm (fails)

One person builds one strategy from start to finish, alone.

- Data, features, model, backtest all done by one quant
- No specialization, no checks
- Overfits to noise, breaks live

Case: LTCM (1998) - one model, huge leverage, blew up in months.

Meta-Strategy Paradigm (works)

A team runs a production line, each stage a specialist.

- Data team, feature team, modeling team, execution team
- Many weak signals combined
- Rigorous validation at each step

Case: Renaissance Medallion - decades of consistent returns.

Why Markets Fight Back

Finance is not like recognizing cats — here is why it's harder

What makes it hard

- Tiny signal, huge noise: prices are mostly random.
- The rules change: a pattern that worked last year can vanish.
- Everyone is competing away the easy patterns.

How we cope (this whole course)

- Sample data smartly (Week 2) and define the question carefully (Week 3).
- Test honestly to avoid fooling ourselves (Week 10).
- Prefer simple, robust models.

A cat is always a cat — a market pattern may quietly stop being true.

The Numbers That Humble Us

Four numbers every financial ML practitioner carries around

< 1%

Typical R^2 of a good daily return model — the signal is tiny

1 : 1,000

Fraud cases per legitimate transactions — true labels are rare

55%

A directional hit rate that already makes a strategy elite

months

How fast a published pattern can decay once everyone trades it

Low signal, rare labels, adaptive markets — exactly why process beats genius.

The Workflow We Follow All Semester

Split, train, validate, test — then deploy. This map reappears every week.

The Machine Learning Workflow

Split your data:



The full pipeline:

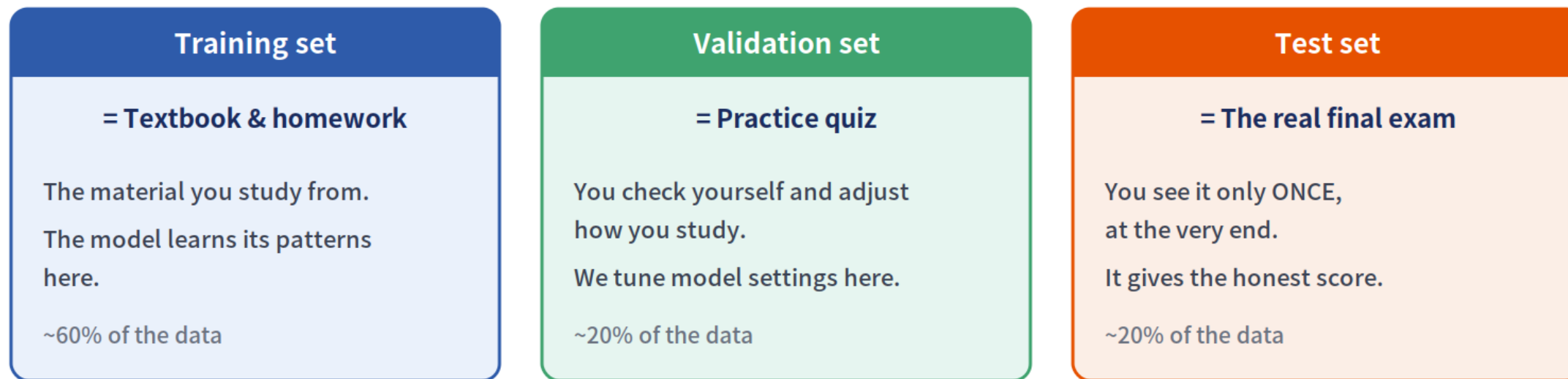


Golden rule: the test set is touched only ONCE, at the very end. Peeking at it inflates your score and fools you.

Train / Validation / Test = Studying for an Exam

Peeking at the test set while training is the #1 beginner mistake — it fakes a great score.

Train / Validation / Test — Like Studying for an Exam



Golden rule: peeking at the final exam while studying = cheating. Your real grade would be a lie.

For Financial Data: Never Shuffle Time

Random splits are fine for cat photos — they are a lie for time series

Why shuffling lies

- A random split mixes future days into the training set.
- The model quietly 'sees tomorrow' while learning — information leaks backward.
- Backtest scores look brilliant; live results collapse.

How we split instead

- Keep chronological order: train on the past, test on the future.
- Walk forward: retrain as time moves, always predicting ahead (Week 10).
- Leave a gap (embargo) so overlapping information can't leak.

In finance the test set is always the future — treat it that way from day one.

Three Real Problems We Will Solve

Each starts simple, then grows more powerful over the weeks

1 · Product Recommendation

suggest the right product

Start simple: 'customers like you also chose...'. Later: neural networks, graphs, even LLMs.

Weeks 5 · 9 · 14

2 · Fraud Detection

catch bad transactions live

Start simple: 'flag anything over \$5,000'. Hard part: only ~1 in 1,000 is fraud. Later: anomaly detection, boosting.

Weeks 6 · 8

3 · AI Chatbot

answer customers automatically

Start simple: keyword matching ('balance' → show balance). Later: language models + retrieval (RAG).

Weeks 14 · 15

The pattern of the course: Traditional → ML → Deep Learning → Frontier.

Getting the Data — One Notebook Does It All

No installs to fear: open it in your browser and press play

Your companion notebook

- Download FML_Data_Setup_Guide.ipynb.
- Open it in Google Colab — free, in the browser.
- Run Setup once, then your week's section.

Two free data sources

- yfinance — real prices, no signup (005930.KS = Samsung).
- OpenML & FRED — credit / macro data, no signup.
- Offline backup — simulated data if needed.

[CODE] FML_Data_Setup_Guide.ipynb · Week 1 loader (§ 2)

```
import yfinance as yf
px = yf.download('AAPL', period='2y', auto_adjust=True,
                 multi_level_index=False, progress=False)['Close']
ret = px.pct_change().dropna()      # daily returns, ready for Lab 1
```

Hands-On Lab 1 — Are Returns a Nice Bell Curve?

Download real prices and look at the shape of daily moves

[CODE] colab_lab1.py · data from setup guide § 2

```
!pip install yfinance scipy -q
import yfinance as yf
import numpy as np, matplotlib.pyplot as plt
from scipy import stats

px = yf.download('AAPL', period='2y', auto_adjust=True,
                 multi_level_index=False, progress=False)['Close']
ret = px.pct_change().dropna()

print('skew      :', stats.skew(ret))
print('kurtosis:', stats.kurtosis(ret))  # >0 = fat tails

fig, ax = plt.subplots(1, 2, figsize=(11, 4))
ax[0].hist(ret, bins=60); ax[0].set_title('Daily returns')
stats.probplot(ret, dist='norm', plot=ax[1]); plt.show()
```

Your task (step by step)

- Pick a ticker (AAPL, or 005930.KS).
- Download ~2 years of prices.
- Turn prices into daily returns.
- Draw a histogram + a Q-Q plot.
- Compute skew and kurtosis.
- Q: are the tails fatter than a bell curve?

Hands-On Lab 2 — Watch Overfitting Happen

See Student A (the memorizer) appear right before your eyes

[CODE] colab_lab2.py · data from setup guide § 2(b)

```
import numpy as np, matplotlib.pyplot as plt

np.random.seed(0)
x = np.linspace(0, 1, 30)
y = np.sin(2*np.pi*x) + np.random.normal(0, 0.2, 30)

xs = np.linspace(0, 1, 200)
for d in [1, 4, 15]:          # simple / good / crazy
    coef = np.polyfit(x, y, d)
    plt.plot(xs, np.polyval(coef, xs), label=f'degree {d}')

plt.scatter(x, y, color='black', s=20)
plt.legend(); plt.ylim(-2, 2); plt.show()
```

Your task (step by step)

- Make wavy data with a little noise.
- Fit 3 curves: deg 1 / 4 / 15.
- Plot all three over the dots.
- Watch deg-15 wiggle through every point.
- Q: which curve would you trust on new data?

Hands-On Lab 3 — Claude as Your Coding Copilot

AI is allowed in this course — it explains and extends; you understand and verify

[INPUT] Prompt · Claude (paste after your Lab 1 code)

You are my machine-learning tutor.

- 1) Take my Lab 1 code above and explain every line in one plain-English sentence each.
- 2) Modify it to download 005930.KS (Samsung) as well, and print skew and excess kurtosis for BOTH tickers in one small comparison table.
- 3) Tell me which ticker has fatter tails, and explain in 3 sentences why fat tails matter for a risk manager, using this week's bias/variance/noise idea.

Do not use any library we have not seen in class.

House rules for AI use

- Allowed: explain, debug, extend code you wrote.
- Required: run it, read it, defend every line.
- Forbidden: submitting code you cannot explain.
- Tip: ask 'why', not just 'do' — that is where you learn.

Comprehension Check

Try each question first — the answer is revealed below

Q1 Predicting loan default (1/0) from labeled history is:

- A** Unsupervised learning
- B** Supervised learning ✓
- C** Reinforcement learning
- D** No learning at all

Answer: We have the answer key (default yes/no), so it's supervised — learning with labels.

Q2 A degree-15 curve that hits every training point most likely has:

- A** High bias, low variance
- B** Low bias, high variance ✓
- C** Low bias, low variance
- D** Zero total error

Answer: It memorized the data (Student A) — it wobbles on new data: high variance = overfitting.

Comprehension Check — Round Two

Two more — these cover the new slides

Q3 You group customers by spending patterns, with no labels at all. This is:

- A** Supervised learning
- B** **Unsupervised learning** ✓
- C** Reinforcement learning
- D** Regression

Answer: No answer key is given — the algorithm discovers the groups itself: unsupervised.

Q4 Your model: train MSE = 0.0, test MSE = 9.0. The gap mostly reflects:

- A** High bias (underfitting)
- B** Irreducible noise
- C** **High variance (overfitting)** ✓
- D** A great model

Answer: Perfect on training, terrible on new data — the train-test gap is the fingerprint of overfitting.

Week 1 — Three Things to Remember

Before next class: finish Labs 1–3 and bring one question about your return histogram

1

ML finds patterns from examples

Three flavors: flashcards (supervised), Lego piles (unsupervised), a video game (reinforcement).

2

Every model balances bias and variance

Total error = Bias² + Variance + Noise. Don't be Student A — memorizing is overfitting.

3

In finance, process beats genius

Markets are noisy and change. Build a careful pipeline (Medallion), not one fragile bet (LTCM).

Next week — Financial Data: bars, CUSUM filters, and why the clock is a bad ruler.